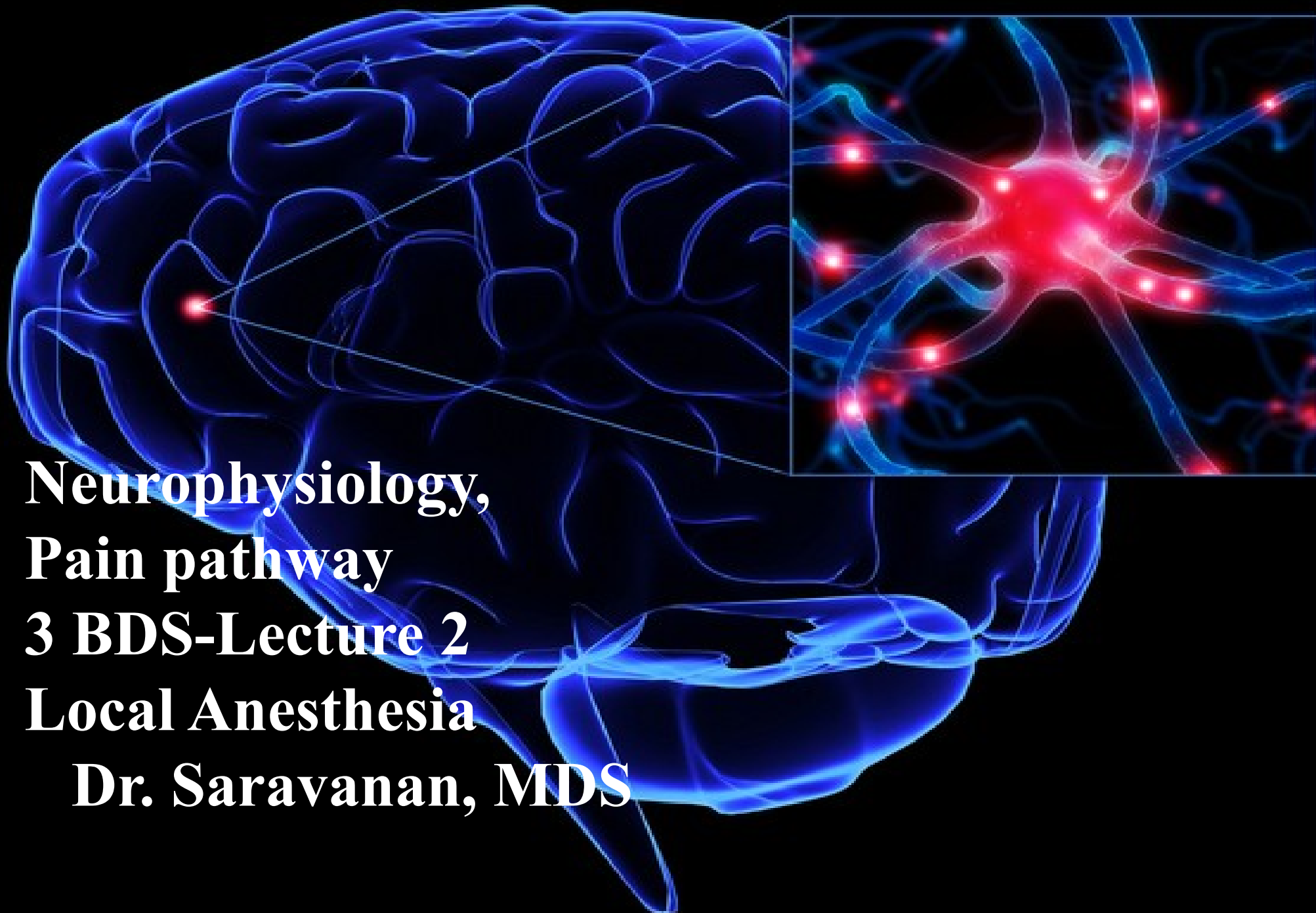
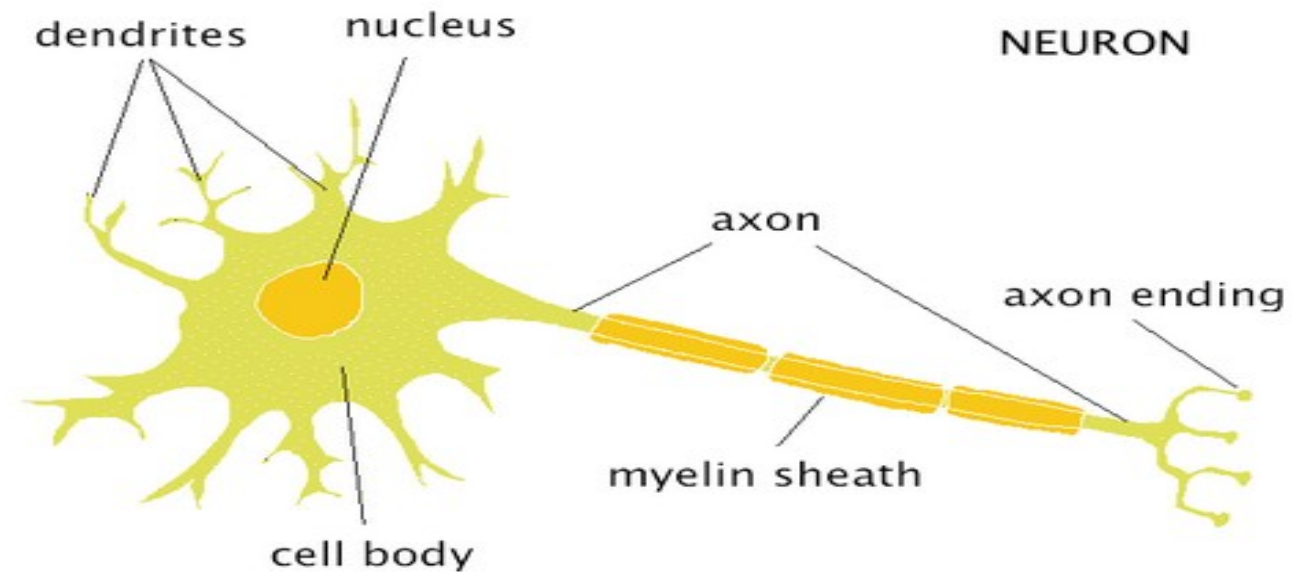




**Neurophysiology,
Pain pathway
3 BDS-Lecture 2
Local Anesthesia
Dr. Saravanan, MDS**

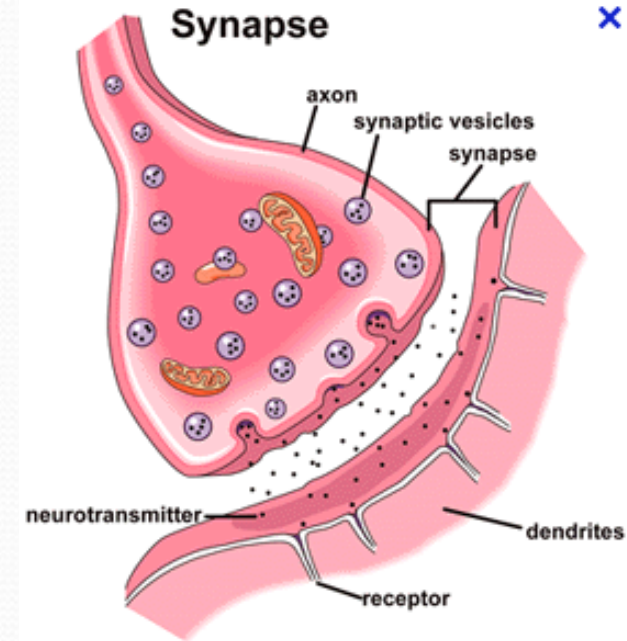
“The Neuron”



- The neuron is the functional unit of the nervous system, it is specialized to transmit electrical nerve **impulses**, thus carry information from one part of the body to another
- The cell body contain the nucleus and has several processes (dendrites) from which impulse enter
- A longer process, the **AXON**, extends outwards and carries impulses to the cell body.
- The point of contact of one neuron with another is known a a **“Synapse”**

synapse

- a **synapse** is a structure that permits a neuron to pass an chemical signal to another neuron
- In a synapse the presynaptic neuron releases a chemical called a **neurotransmitter** that binds to receptors located in the postsynaptic cell, usually embedded in the plasma membrane.
- The neurotransmitter changes the postsynaptic cell's excitability to cause an action potential in the postsynaptic cell and a continuation of the "message."

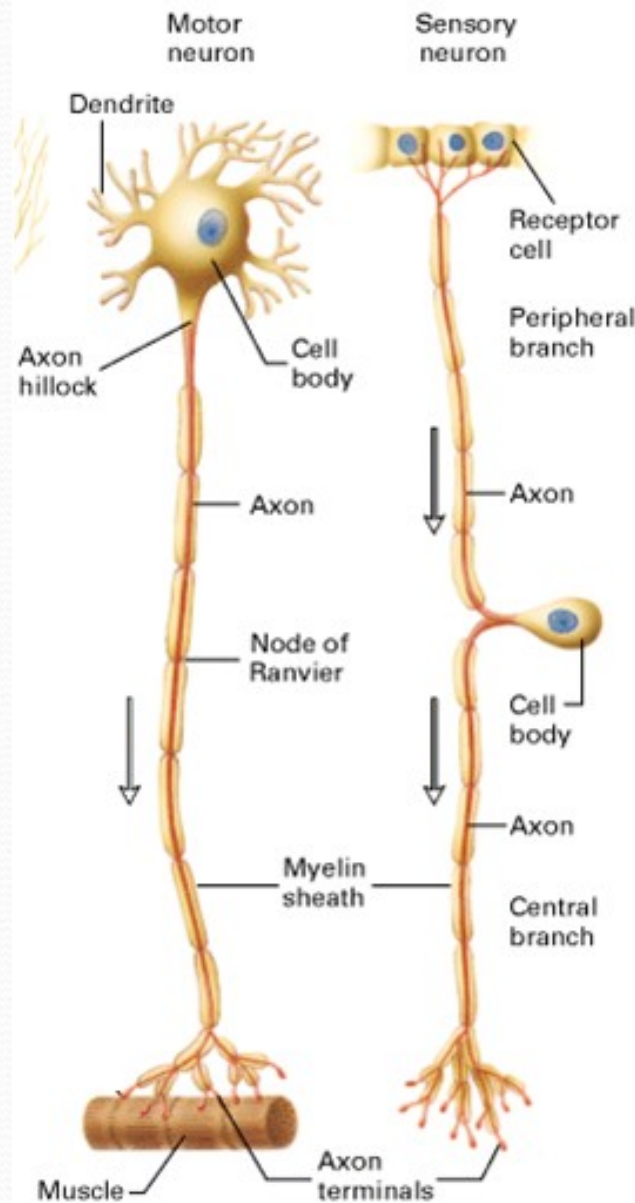


Types of Neuron

Motor neuron

(efferent)

- Nerve cells conduct impulses from CNS to periphery.
- The cell body of motor neuron is interposed between axon and dendrites.

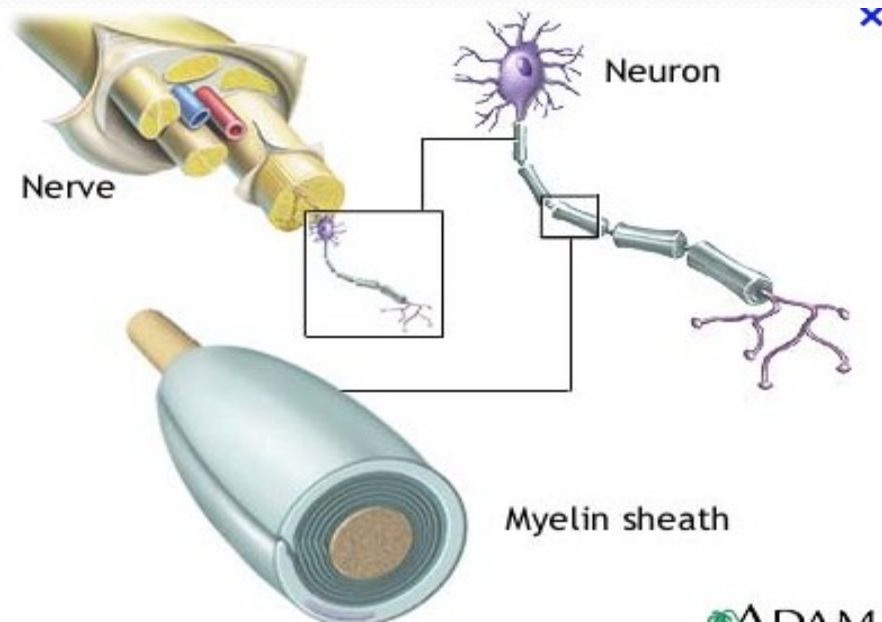


Sensory neuron

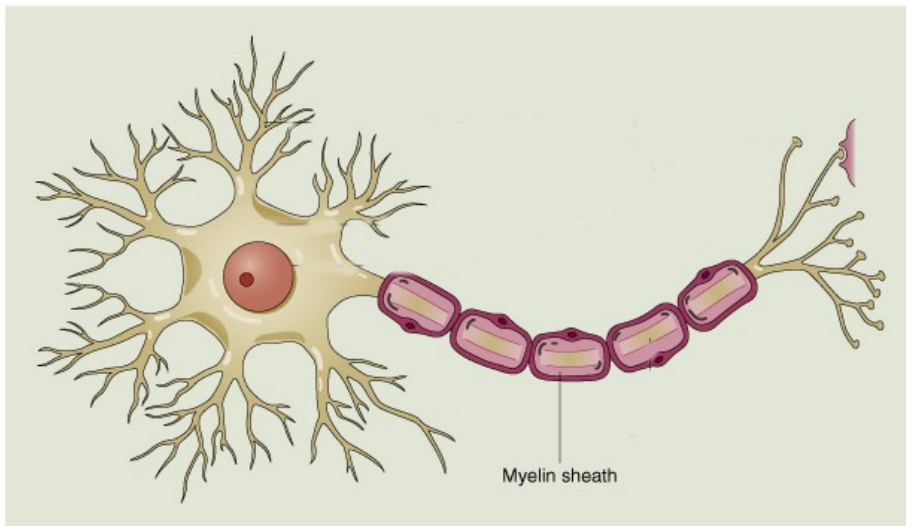
(afferent)

- activated by sensory input (vision, touch, hearing, pain etc.), and send projections into the CNS that convey sensory information to the brain or spinal cord.
- The cell body is located at a distance from the axon.

Myelinated/unmyelinated nerves



- Some nerves are covered by an **insulating lipid layer of myelin**.
- Myelinated neurons are enclosed in spirally wrapped layers of lipoprotein myelin sheaths, which are specialized form of **schwann cells**.
- There are gaps between two adjoining myelin sheaths, they are called “**nodes of Ranvier**”

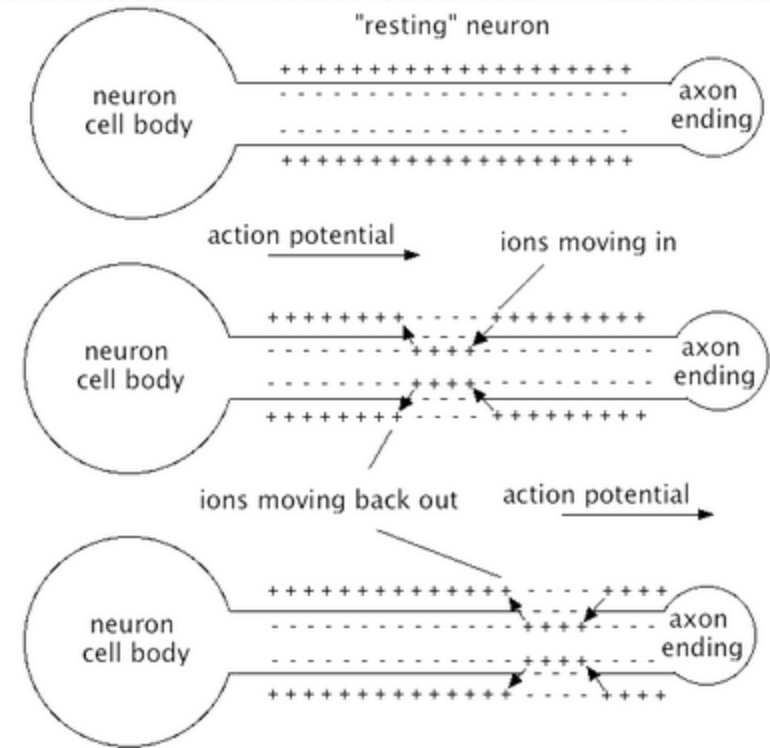


The action potential

When chemicals contact the surface of a neuron, Action potentials are caused by an exchange of **ions** across the neuron membrane .

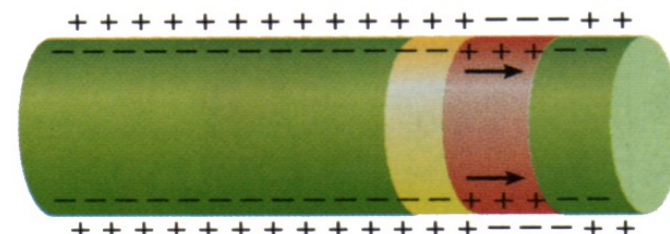
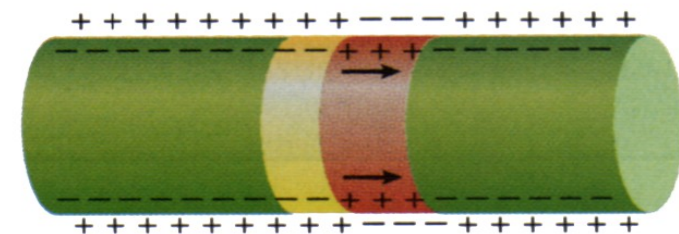
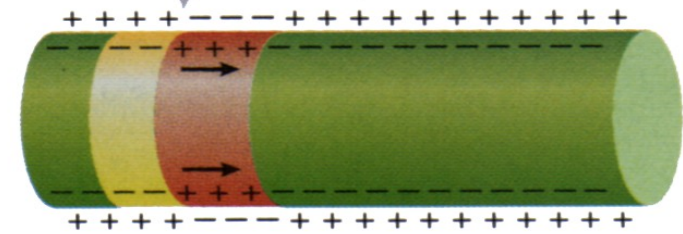
Chemicals in the body are "electrically-charged" -- when they have an electrical charge, they are called **ions**. The important ions in the nervous system are sodium and potassium (both have 1 positive charge, +), calcium (has 2 positive charges ++) and chloride (has a negative charge, -).

A stimulus first causes **sodium channels** to open. Because there are many more sodium ions on the outside, and the inside of the neuron is negative relative to the outside, sodium ions rush into the neuron. Remember, sodium has a positive charge, so the neuron becomes **more positive and becomes depolarized**



Conduction of Nerve Impulse in an Unmyelinated Fibers

-  Action potential in progress
-  Refractory membrane
-  Excitable membrane



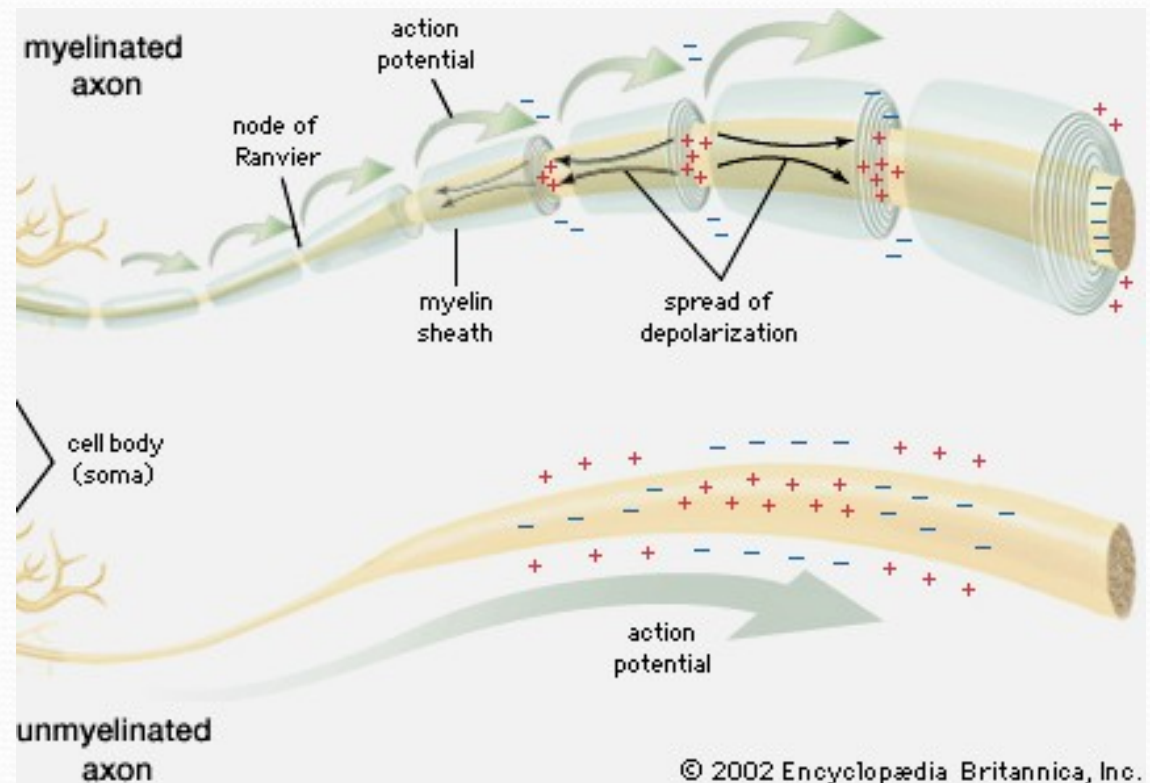
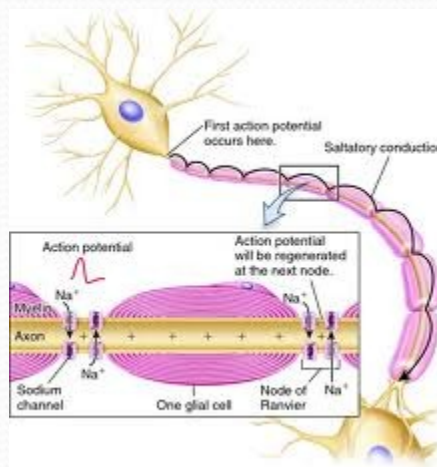
- In unmyelinated neurons, the action potential travels in a slow forward creeping process
- A region of membrane is in its refractory period and prevents the impulses from going backward

Action potential propagation in myelinated fibres

Impulse conduction in myelinated nerves occurs by electrical impulse leaps from node to node.

This process is termed Saltatory conduction.

This form of conduction is much faster.



Types of nerve fibres

Type A

- Large myelinated fibers
- 3-20 Micron in thickness
- Conduct fast localized pain
- Rate of conduction is 100 meter/second

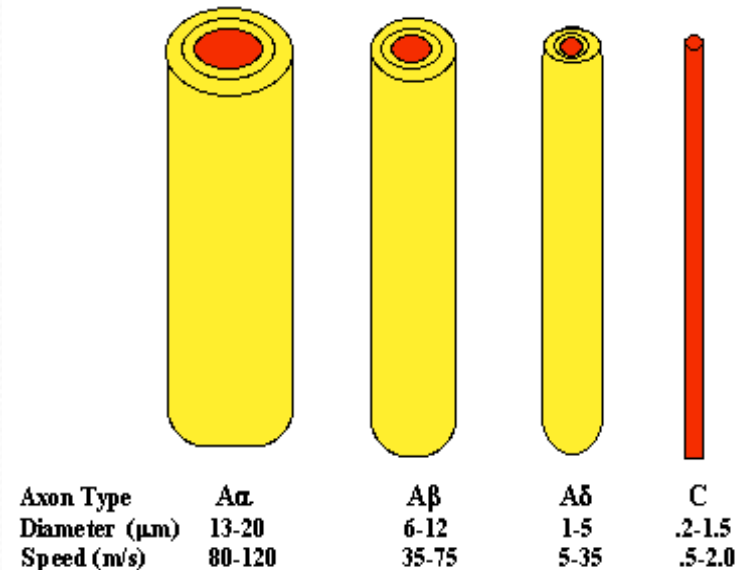
Type B

- Up to 3 micron in diameter
- Rate of conduction is 3-14 meter/second

Type C

- Small unmyelinated fibers
- 0.5-1.0 micron in diameter
- Conduct slower secondary pain
- Rate of conduction is 0.5-2.0 meter/second

Primary Afferent Axons



information about the different primary afferent axons:

A-alpha nerve fibers carry information related to proprioception (muscle sense).

A-beta nerve fibers carry information related to touch.

A-delta nerve fibers carry information related to pain and temperature.

C-nerve fibers carry information related to pain, temperature and itch.

Pain and Pain Pathway

Pain is an unpleasant sensation created by noxious stimuli and mediated along specific nerve pathway to the central nervous system where it is interrupted as pain

Pain Is a Protective Mechanism.
Pain occurs whenever any tissues are being damaged, and it causes the individual to react to remove the pain stimulus





Pain has been classified into two major types:

fast pain
slow pain.

Fast pain

- *Fast pain* is felt within about 0.1 second after a pain stimulus
- Fast pain is also described by many alternative names, such as *sharp pain, pricking pain, acute pain, and electric pain.*
- *This type of pain is felt when a needle is stuck into the skin, when the skin is cut with a knife, or when the skin is acutely burned. Fast-sharp pain is not felt in most deeper tissues of the body.*
- It is better localized
- The type A-delta pain fibres transmit mainly mechanical and acute thermal pain

Slow pain

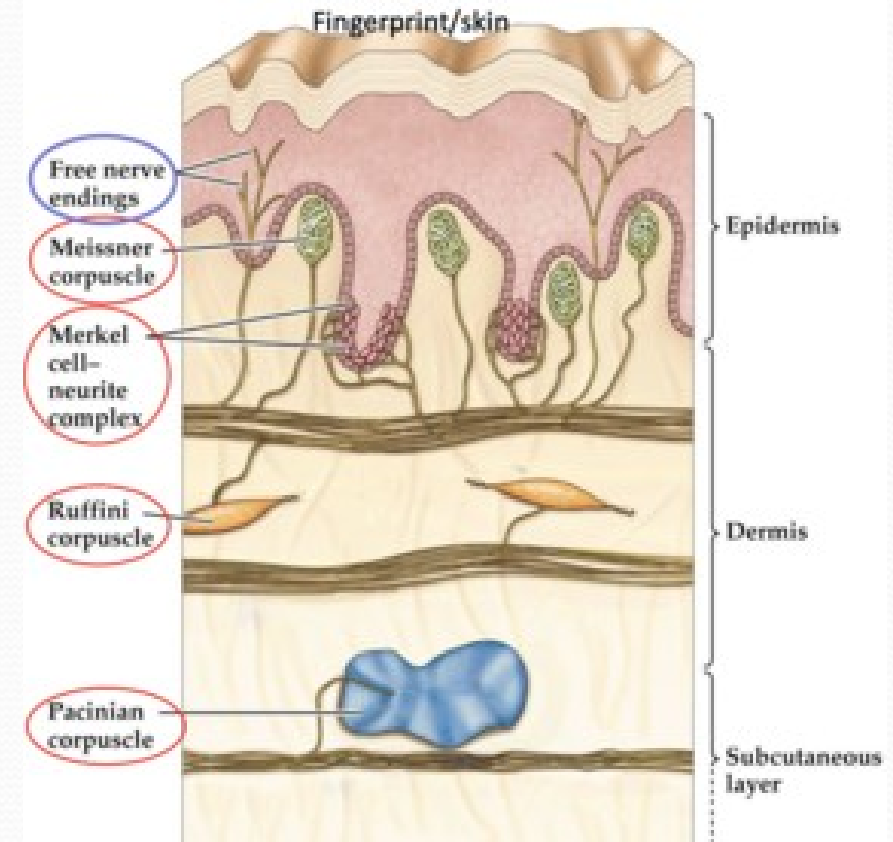
- slow pain begins only after 1 second or more and then increases slowly over many seconds and sometimes even minutes
- Slow pain also called such as *slow burning pain, aching pain, throbbing pain, and chronic pain*. This type of pain is usually associated with **tissue destruction**. It can lead to *prolonged, unbearable suffering*. It can occur both in the skin and in almost any deep tissue or organ.
- not localized
- Slow pain mainly carried in the peripheral slow-chronic **type-C pain** fibers

Pain Receptors

Are called Free Nerve Endings

Also called “Nociceptors”

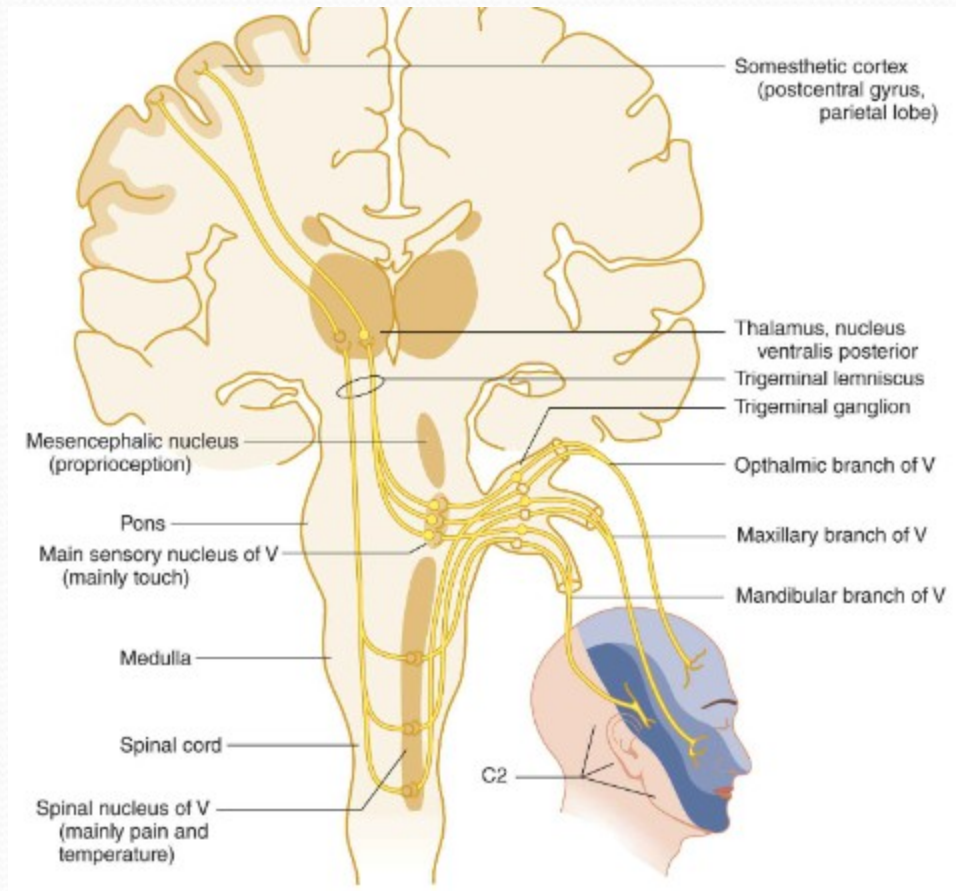
- nerve endings that have no special association with specific cells or tissues
- They are most abundant in epithelial and connective tissues
- Mostly unmyelinated
- In the form of loops or long hair like network



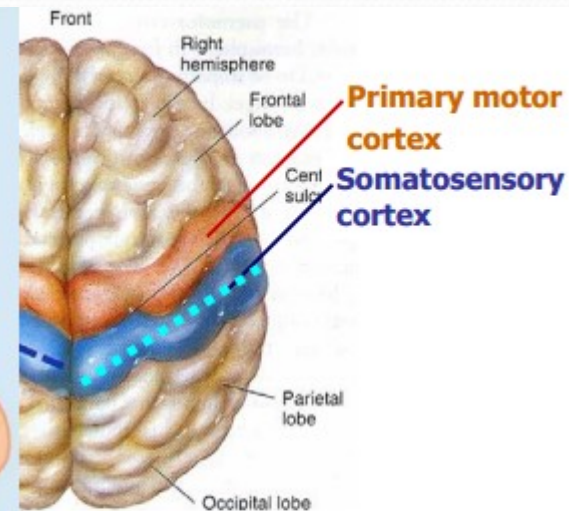
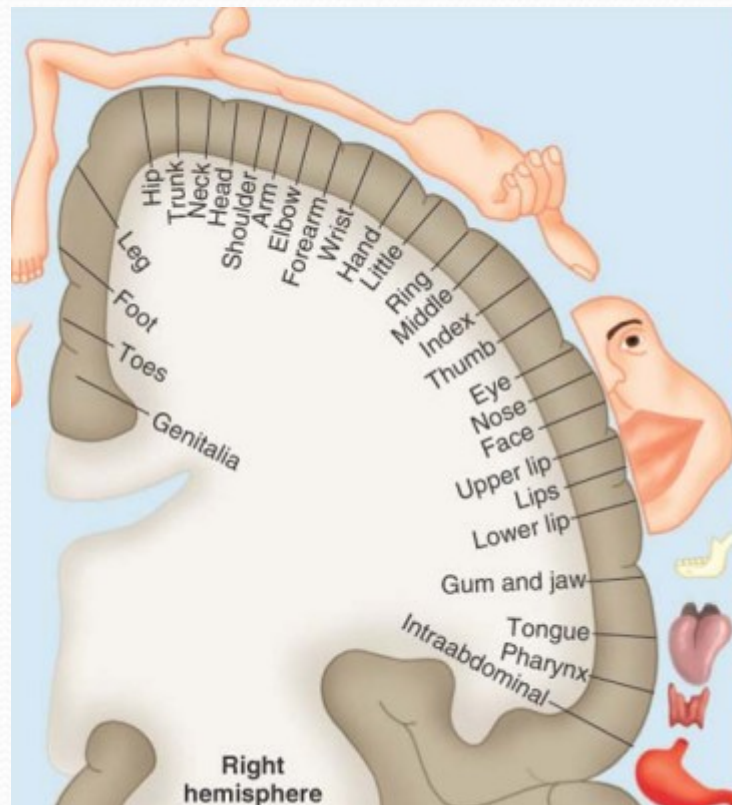
Orofacial Pain Pathway

Somatic inputs from the face and oral structures are carried by way of the fifth cranial nerve, the **trigeminal nerve**.

Axons of primary (1°) sensory neurons enter the brain stem, travel in the descending (spinal) tract of V, and terminate in the descending (spinal) nucleus of V



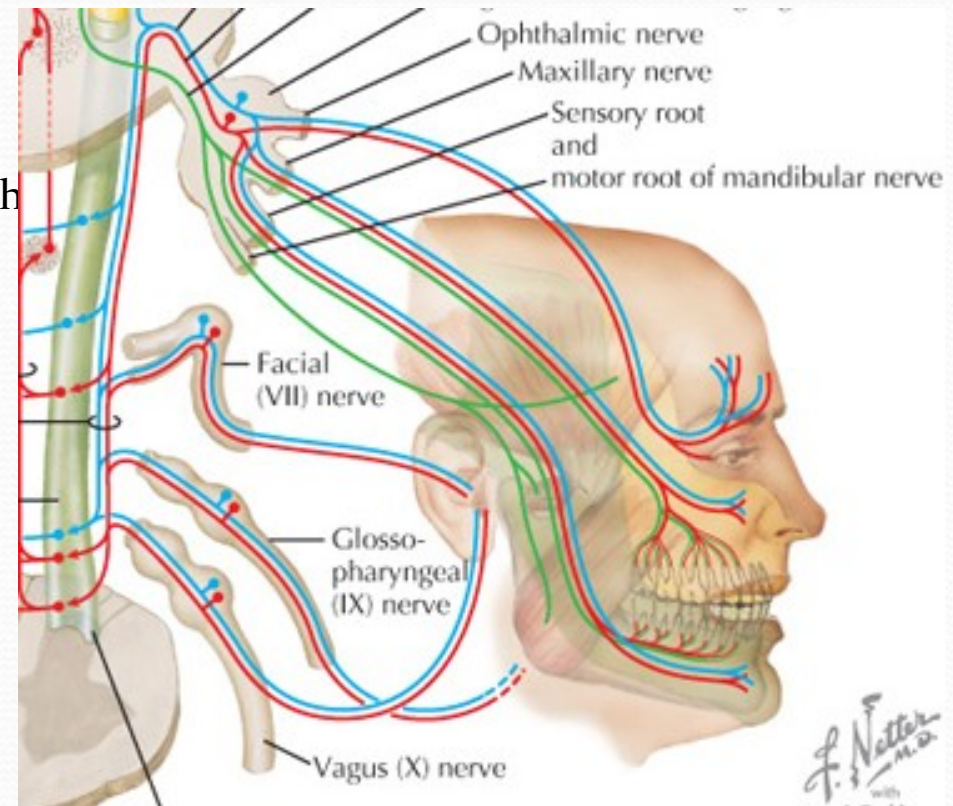
Primary Sensory Cerebral Cortex



Somatosensory areas in the cortex of the brain are anatomically organized in relation to the source of information, with

Other sensory pathway

- Nociceptive impulses from the face and mouth may be mediated centrally by way of afferent neurons that pass through the 7th, 9th and 10th cranial nerves
- axons of the **geniculate ganglion (VII)** and **jugular ganglion (X)** supply a small zone of the external ear. Axons of the **petrosal ganglion (IX)** supply general sensation to the posterior oral cavity and pharynx.



HOW TO MEASURE PAIN

- Pain measurement is a complex and controversial psycho physiologic subject
- There is no simple method of measuring pain. Pain intensity can be measured by using ratings such as

VISUAL ANALOG SCALE (VAS).

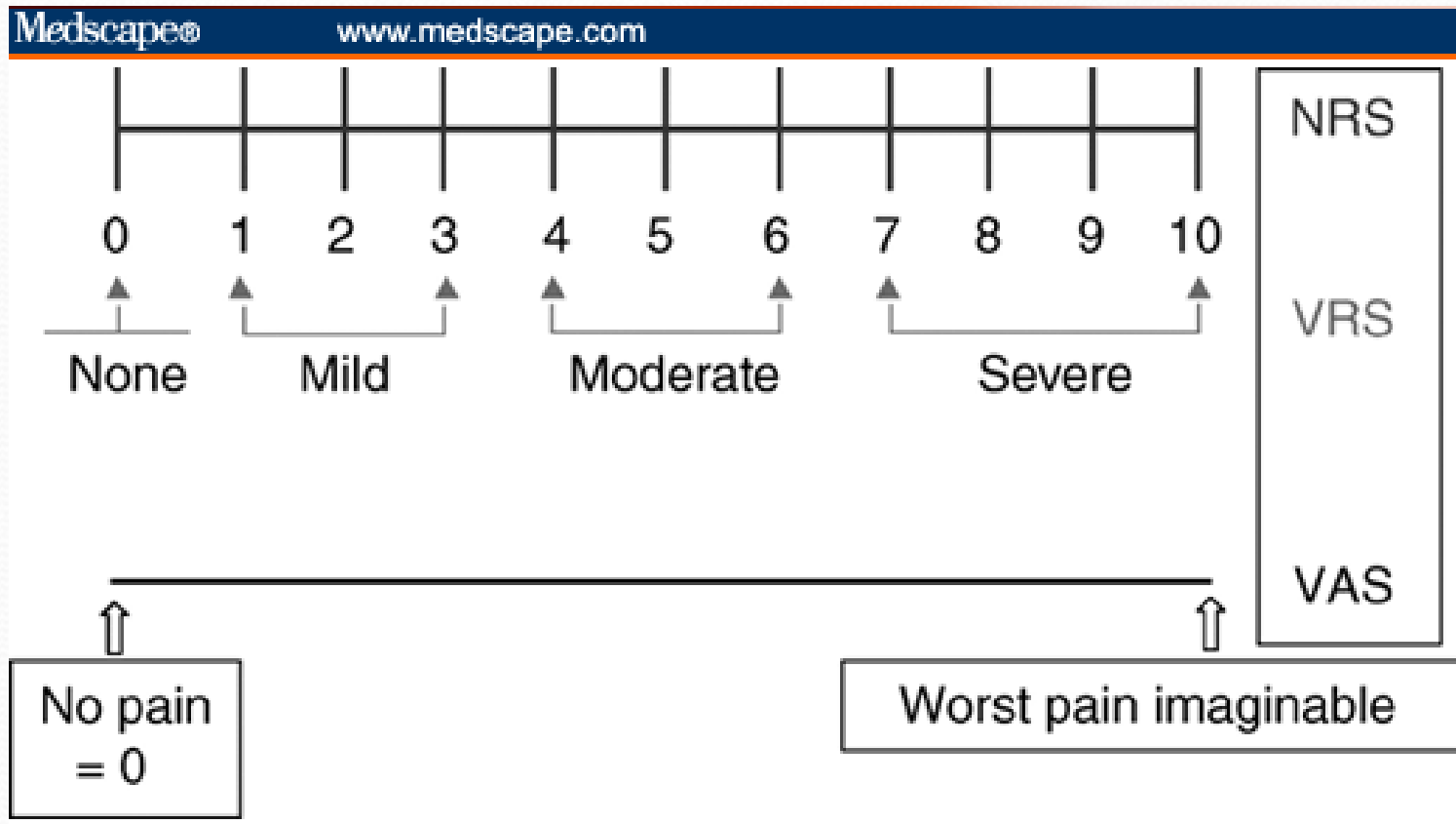
A VAS consists of a 10cm line on which

0 → no pain 10 → pain as bad as it can get

The VAS are sensitive to **treatment effects**, can be **incorporated into pain diaries** and can be used with **children**.

Disadvantage: the multidimensional aspects of pain are not well measured

Visual analog scale (VAS)





Thank You